

## C2-05 Mini-Lab - Weekly Reporting Pipeline

Integrated automation practice after PYA1-PYA11

### SCENARIO

Operations receives approved Apr-Jun sales files plus a dirty customer master. Build a callable reporting pipeline that can run from a clean process, reject unsafe input, publish a regional output only after critical checks, and leave enough evidence for another analyst to reproduce the run.

### MUST DEMONSTRATE

Config-driven paths/period; expected-vs-actual file discovery; schema/types; source\_file lineage; master-key uniqueness before merge; unmatched-key evidence; many-to-one merge validation; log + structured validation JSON; Excel/CSV output verification; run manifest; second unchanged rerun; one failure that withholds release.

### PASS SIGNAL

Raw inputs remain unchanged. Second identical run does not duplicate business rows. A critical validation failure prevents final report publication. Log + validation + manifest explain what happened and what produced the released result.

# Mini-Lab - Defense and changed retry

Close protected review before transfer

## 90-SECOND DEFENSE

Explain: What is the input/output grain? Which files were approved? What prevents duplicate customer keys from multiplying rows? What check gates publication? What proves rerun safety? What exact code/config/input/environment evidence lets another analyst reproduce the run?

## CHANGED RETRY

After review, move the project under a different parent folder and switch the reporting period/input directory through config only. Add one new validation rule. Re-run from a clean process without editing transformation logic.

## EVIDENCE

Save script/notebook, config, run.log, validation\_results.json, report output, manifest, expected-vs-actual files, exception rows, first/second-run controls and failure-withheld evidence.